IS6400 Business Data Analytics

Dr. LIU Junming

City University of Hong Kong

Lecture 1

- ❑ **Course Information**
- ❑ Introduction
- ❑ Course Overview
- ❑ Platform Introduction

Reading Material:

Instructor Information

- Instructor:

  Junming Liu, Ph.D. AC3 6-277 (3442-9323)
  junmiliu@cityu.edu.hk

- Co-Instructor and TA:
  - SHUAI Yanqin, [contact removed]
  - QIN Yaxue, [contact removed]
  - Yu Shujing, [contact removed]

Office Hour: Mon. 12:00-14:00, by email appointment

Class Policy

- No Cell Phone in classroom
  - Pay mutual respect to each other
  - Do not disturb your fellow classmates

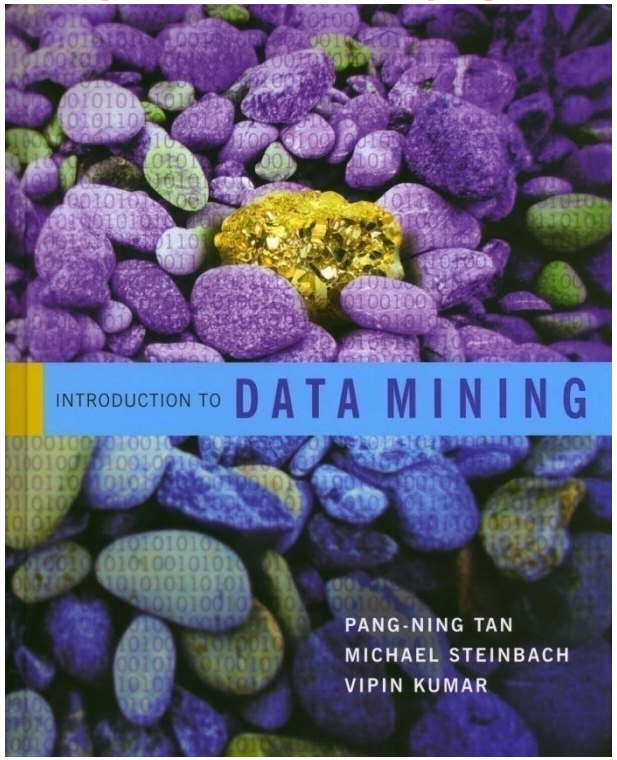


- Attend the classes punctually

- Academic Honesty
  - Plagiarism is a SERIOUS offense at CityU
  - Don’t do it. If you’re in doubt, discuss with your lecturer or tutor

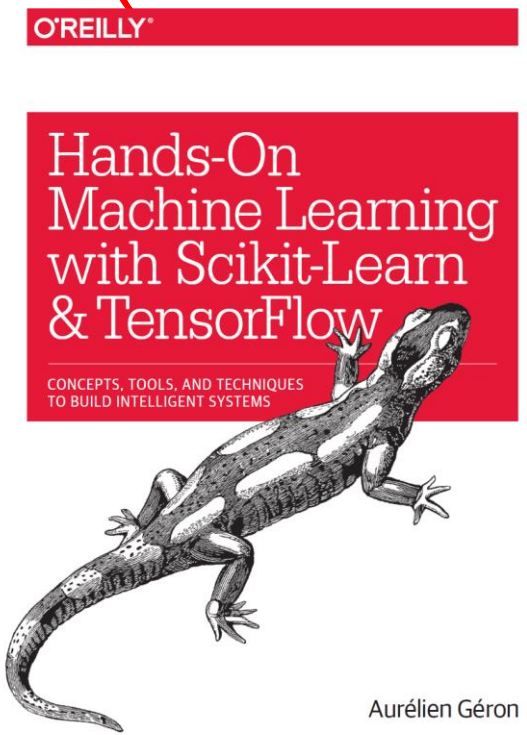
Required Textbook

DM (modeling part)



Introduction to Data Mining
Pang-Ning Tan, Michael Steinbach, Vipin Kumar,
ISBN: 0-321-32136-7, 2005. eBook Link

HM (source codes)



Hands-On Machine Learning with Scikit-Learn & TensorFlow

Course Materials

- Platform: **Anaconda** – The World's Most Popular Python/R Data Science Platform
- https://www.anaconda.com/download

- Library: Data Mining techniques using Python packages
  - http://scikit-learn.org/stable/
  - http://scikit-learn.org/stable/auto_examples/index.html

- Library: TensorFlow
  - https://www.tensorflow.org/

Tentative Teaching Schedule

COURSE SCHEDULE (*TENTATIVE*)

Week#	Lecture	Hands-on Tutorial	Tutor
1	Course Introduction	Software installation	Yaxue Qin
2	Predictive Analytics: Multivariate regression	Regression Analysis	Yaxue Qin
3	Feature Selection and Dimension Reduction	Principle Component Analysis	Yaxue Qin
4	Bisecting KMeans, DBSCAN	Clustering Methods	Yaxue Qin
5	Supervised Learning: Decision Trees, Ensemble Learning	Classification Methods	Yaxue Qin
6	Project Briefing		
7	Time series analysis (1)	Holt's Model	Yanqin Shuai
8	Time series analysis (2)	Exponential smoothing	Yanqin Shuai
9	Deep Learning: ANN and RNN	Nonlinear Time Series	Yanqin Shuai
10	Deep Learning: Image Mining	Image Recognition	Yanqin Shuai
11	Model Assessment Competition: GenAI for BDA	Model Selection	Yanqin Shuai
12&13	Group Project Presentation		

Course Intended Learning Outcome

- Understand the importance of data analytics for business
- Learn the Advanced concepts and techniques of Machine Learning and Deep Learning
- Apply learned techniques for Real-world problems
- Identify appropriate tools and techniques for innovative business solutions

Show something on your cv!

Assessment Criteria

- Participation (5 %)
  - Class attendance and participation, in-class quiz, etc.
- Group Project (30%)
  - Project Briefing (5%)
  - Presentation (15%)
  - Final report (10%)
- Individual Assignments (25 %)
  - 5~10 assignments due next WEEK
- Final Exam (40%) – 2hour (open/closed book)

Attendance

Regular attendance is compulsory. You are not allowed to check your emails, access Web sites not related to the course or work on something that is beyond the scope of this course during the class time.

If you want to finish your assignment quickly?

If you want to complete your project?

If you want to get some extra credits?

Assignments

You may have discussions with your class members, but you must submit your own work. Please be sure to keep a copy of the assignment by yourself in case that there is any problem with your hand-in or you have to use it later this semester.

Submit before the beginning of the class on the specified due day.

Late submissions will be accepted. (-20 points per day overdue)

Course Project

- A course project using any data analytics models using real word data.
- Max 6 students in a group.
- Dataset Available:
  - Kaggle: https://www.kaggle.com/datasets
  - Kaggle: https://www.kaggle.com/competitions
  - UCI: https://archive.ics.uci.edu/ml/index.php
  - Google Dataset: https://datasetsearch.research.google.com/

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- ❑ Platform Introduction

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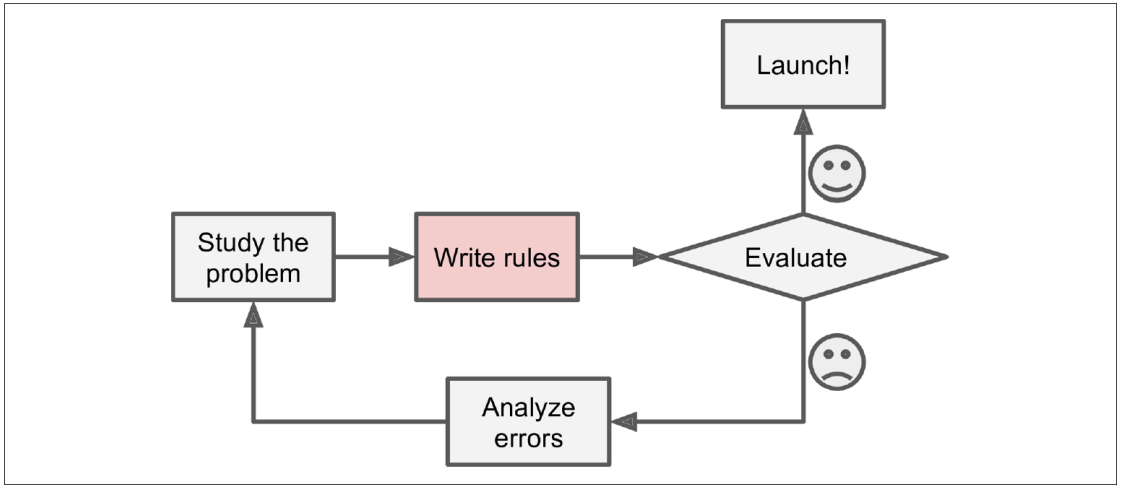
What is Data Analytics?

- **Tell computers to learn from Data**
  - Data Analytics is the process of automatically **discovering useful information in large data repositories**.
  - It is a **technology** that blends traditional data analysis methods with sophisticated algorithms for **processing large volumes of data**.
  - It has also opened exciting opportunities for **exploring and analyzing new types of data** and for analyzing old types of data in new ways.

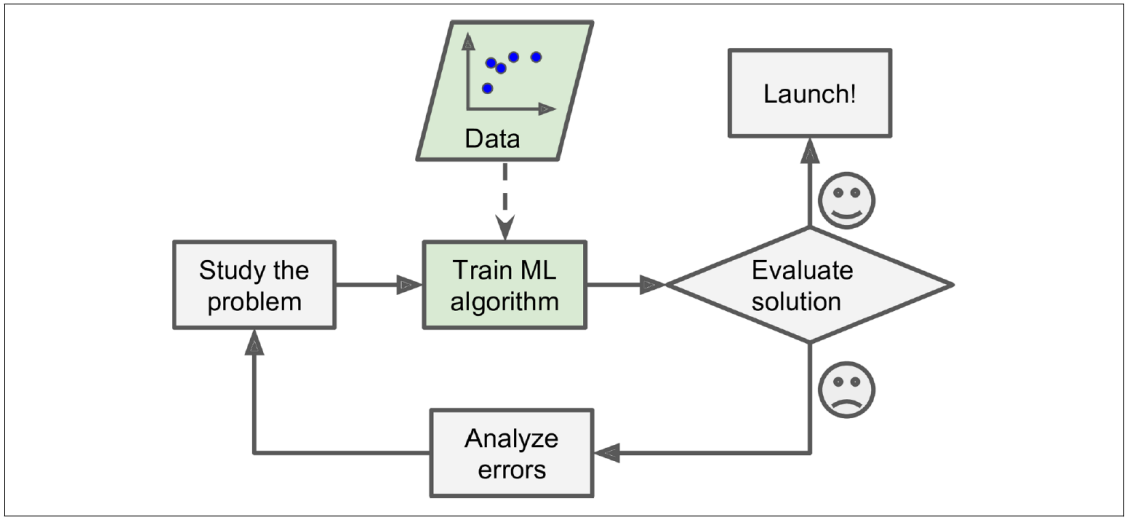
Machine Learning for DA?

Tell computers to learn from Data like a human

Human Learning

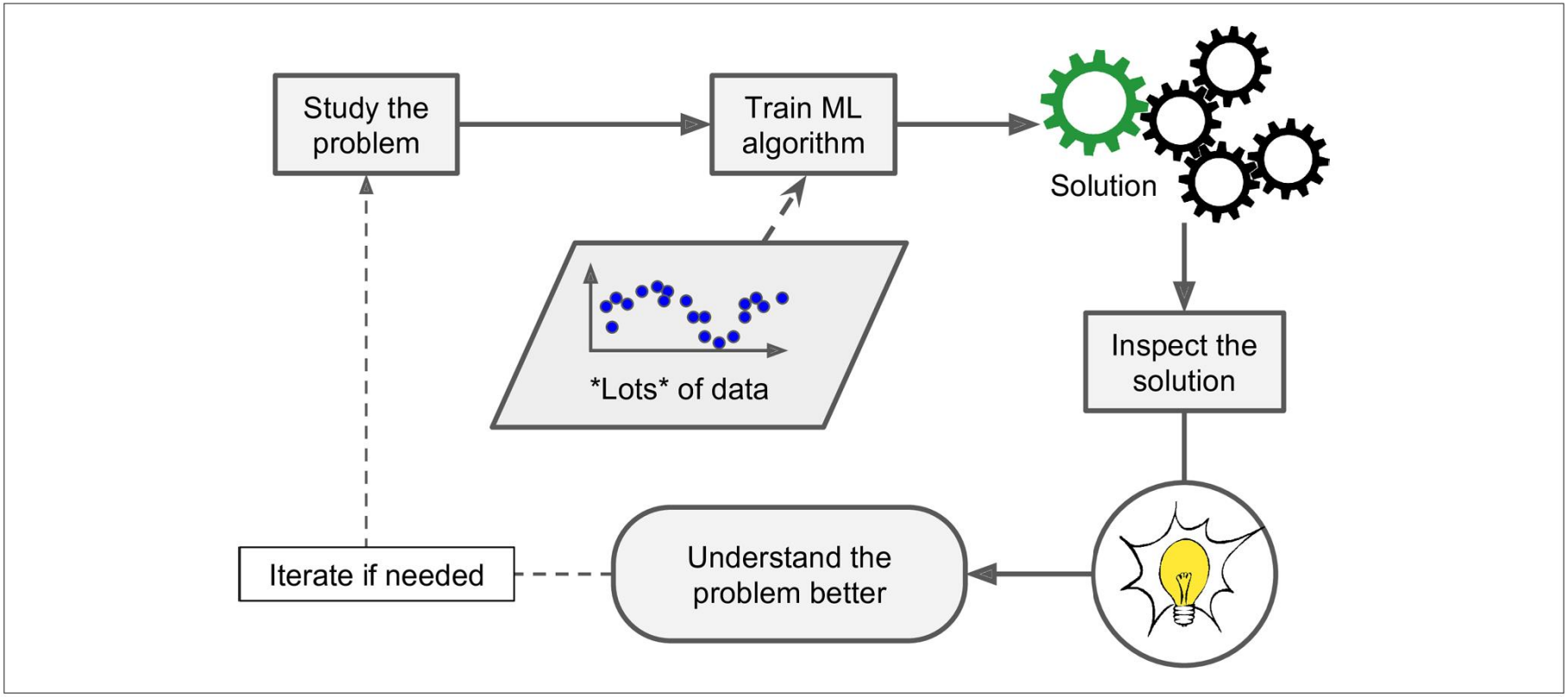


Machine Learning



Why use Machine Learning?

Help a human learn with a machine



Real-world Applications

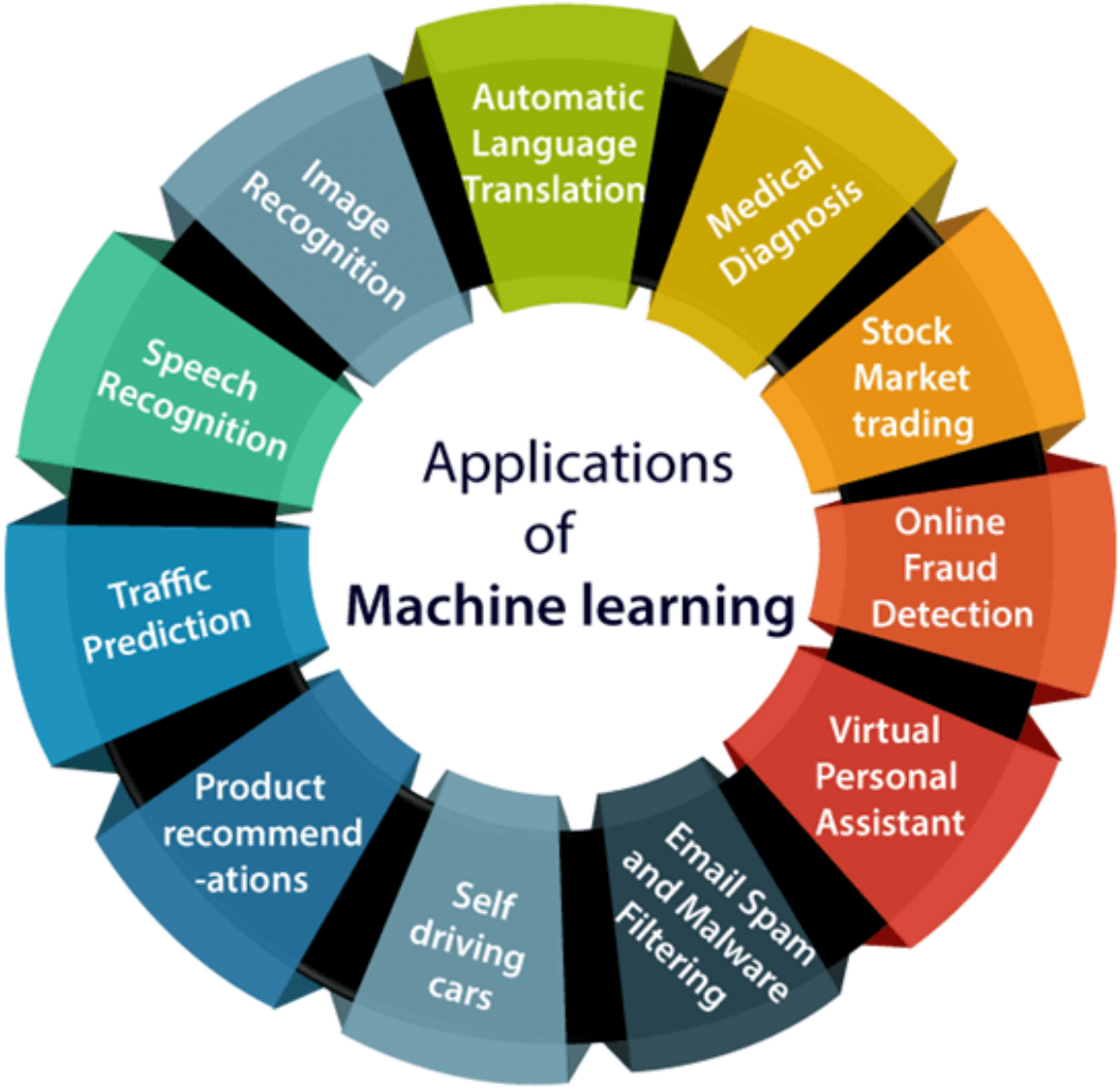


Image recognition – Example

- Image Recognition and Artificial Intelligence



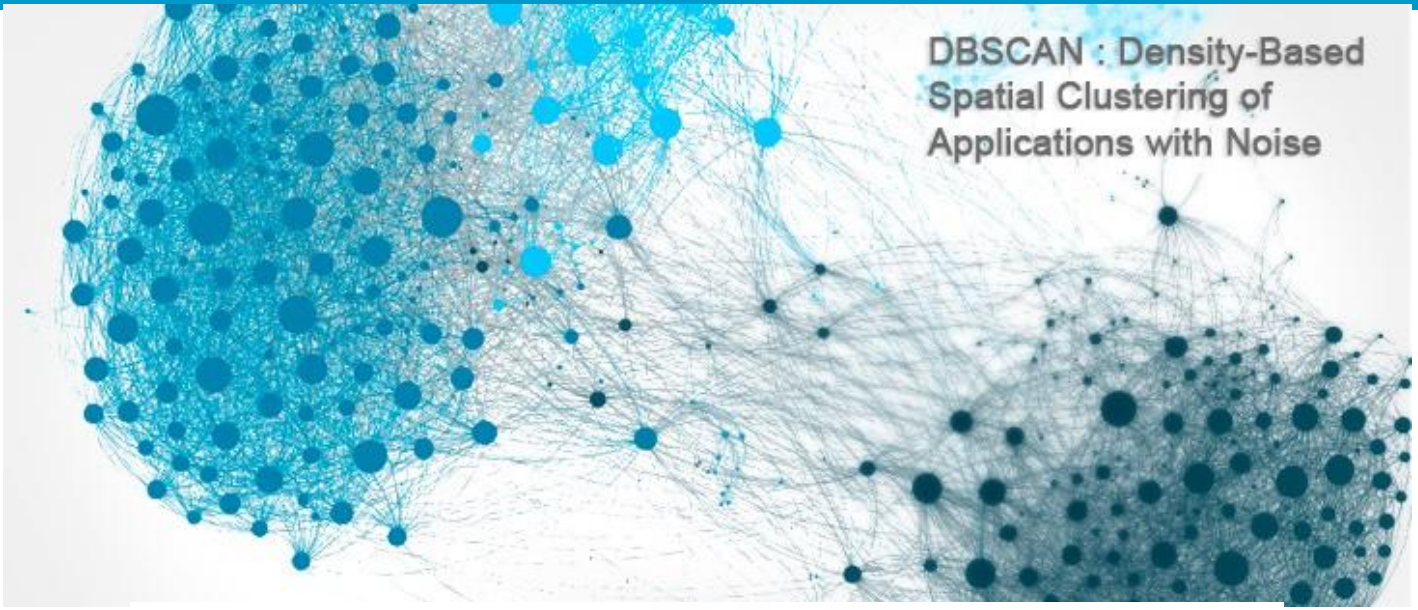


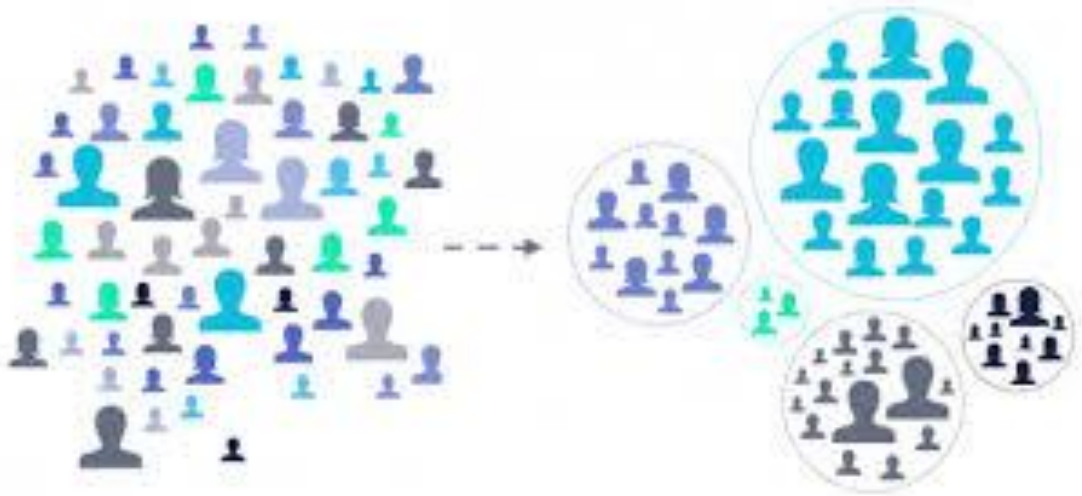
Stock Market – Example

- Stock price forecasting http://theoreti.ca/?p=5032



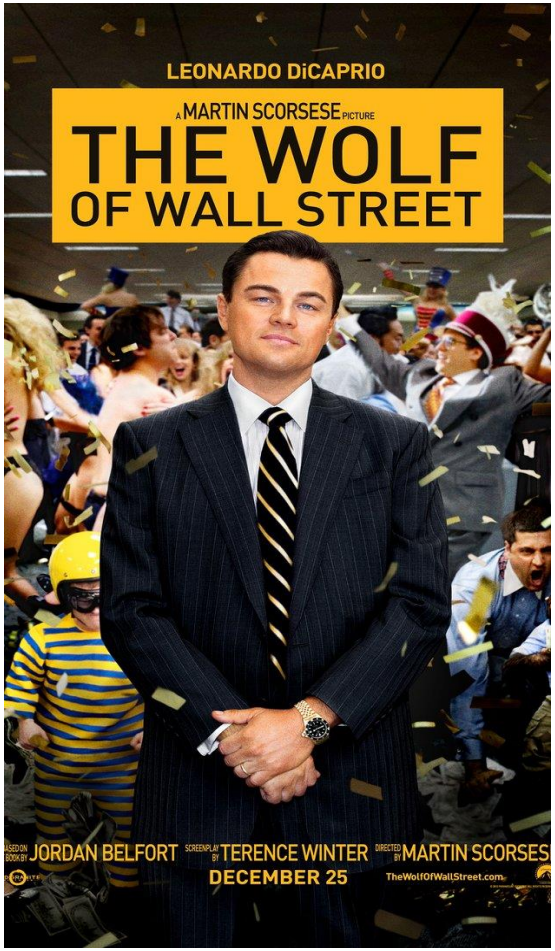
Clustering – Social Network Segmentation





Why Machine Learning for DA?

The Age of Big Data need machine to process data and support decision making





Life in 1990's



Life in 2010's

My Business Applications

Intelligent Production, Replenishment and Transshipment System for Retailers (perishable product and fast fashion)





Redistribute Stocks

Product List:
P1, P2, P3,…



Unbalanced consumption among products in-store

Unbalanced consumption among retailers

Representative Business Clients:

PEACEBIRD MEN

Fast Fashion
4300+ Retail Stores



Dairy Company
1800+ Retail Stores

PEAK

Sport Fashion
1200+ Retail Stores

My Ongoing Applications

Use social media to shape fashion trend

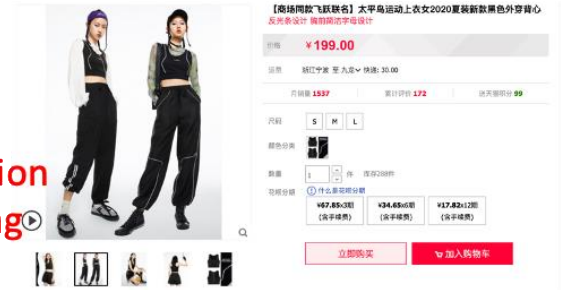




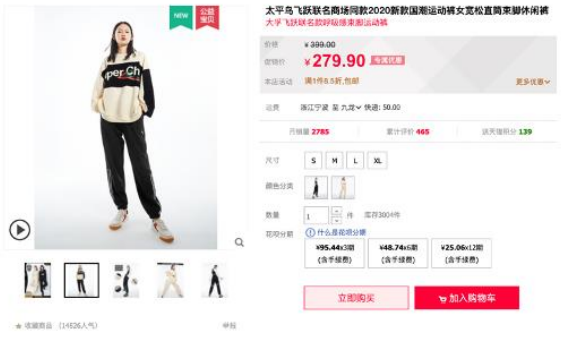


Social Media Activity

Fast Fashion Top Selling







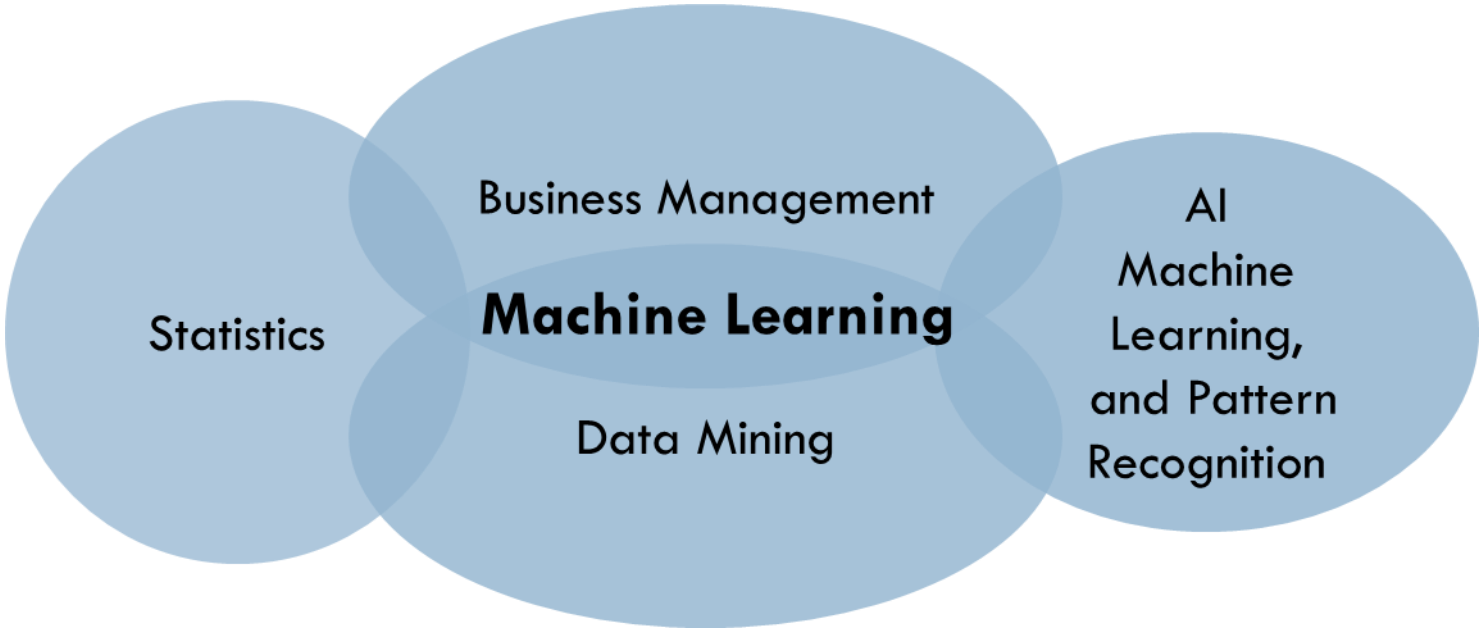
Business Applications

- Pricing
  - setting prices for consumer and industrial goods, government contracts, and maintenance contracts
- Customer segmentation
  - identifying and targeting key customer groups in retail, insurance, and credit card industries
- Merchandising
  - determining brands to buy, quantities, and allocations
- Location
  - finding the best location for bank branches and ATMs, or where to service industrial equipment
- Social Media
  - understand trends and customer perceptions; assist marketing managers and product designers

Machine Learning Applications

- Market analysis
- Risk analysis and management
- Fraud detection and detection of unusual patterns (outliers)
- Text mining (news group, email, documents) and Web mining
- Stream data mining

Interdisciplinary Study

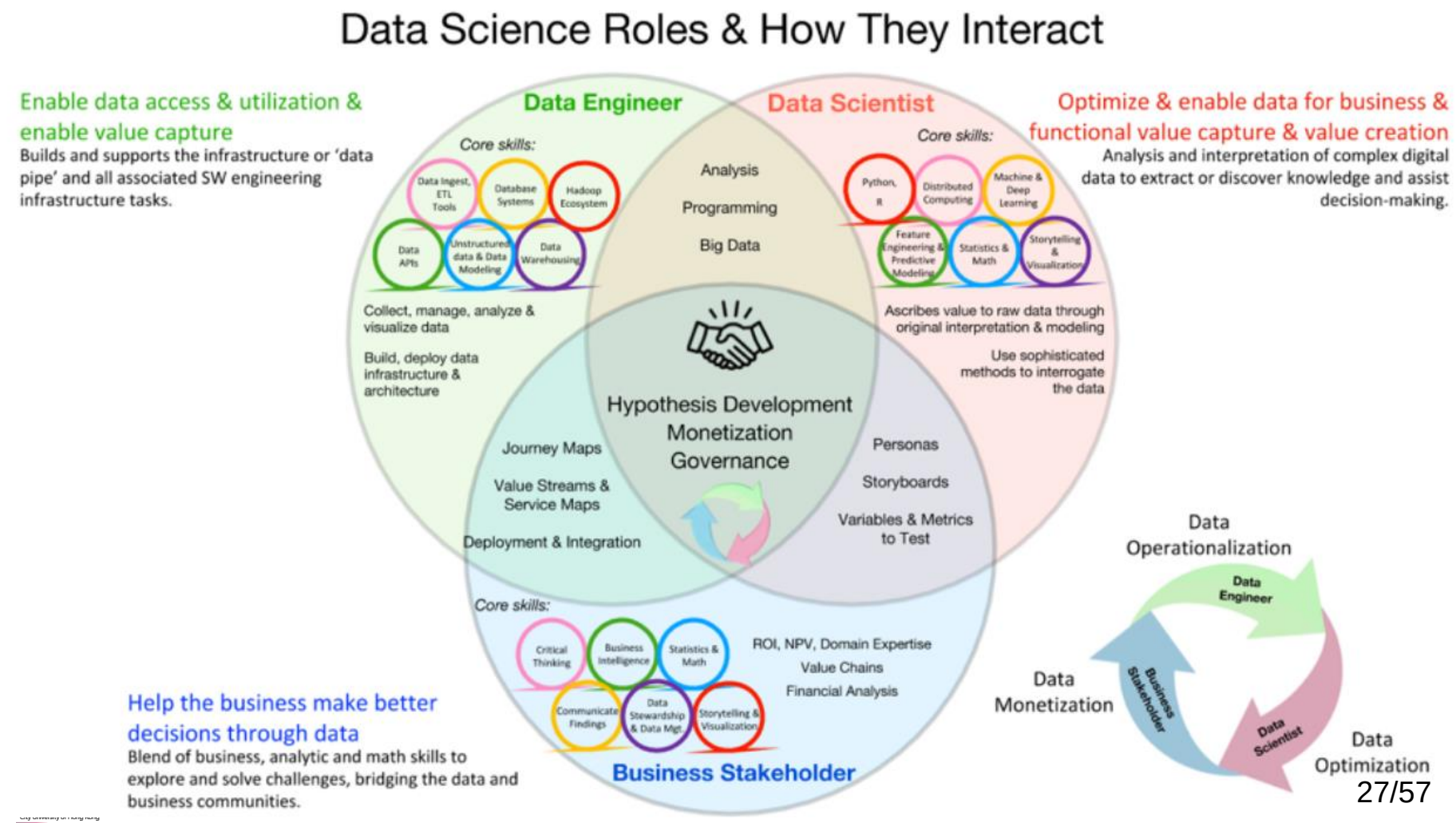


You Play in everyone's sandbox



The best thing about analytics is that it lets you play in everyone's sandbox

Jobs in BDA


Data Science Roles & How They Interact
Enable data access & utilization & enable value capture
Builds and supports the infrastructure or 'data pipe' and all associated SW engineering infrastructure tasks.
Data Engineer
Core skills:
Data Ingest, ETL Tools
Database Systems
Hadoop Ecosystem
Data APIs
Unstructured data & Data Modeling
Data Warehousing
Collect, manage, analyze & visualize data
Build, deploy data infrastructure & architecture
Data Scientist
Core skills:
Python, R
Distributed Computing
Machine & Deep Learning
Feature Engineering & Predictive Modeling
Statistics & Math
Storytelling & Visualization
Ascribes value to raw data through original interpretation & modeling
Use sophisticated methods to interrogate the data
Optimize & enable data for business & functional value capture & value creation
Analysis and interpretation of complex digital data to extract or discover knowledge and assist decision-making.
Analysis
Programming
Big Data
Hypothesis Development
Monetization
Governance
Journey Maps
Value Streams & Service Maps
Deployment & Integration
Personas
Storyboards
Variables & Metrics to Test
Core skills:
Critical Thinking
Business Intelligence
Statistics & Math
Communicate Findings
Data Stewardship & Data Mgt.
Storytelling & Visualization
ROI, NPV, Domain Expertise
Value Chains
Financial Analysis
Business Stakeholder
Help the business make better decisions through data
Blend of business, analytic and math skills to explore and solve challenges, bridging the data and business communities.
Data Operationalization
Data Engineer
Business Stakeholder
Data Scientist
Data Monetization
Data Optimization

Career Future

- Shortage:
  USA alone faces a shortage of 140,000 to 190,000 people with deep analytical skills

- Shortage:
  1.5 million managers to analyze big data and make decisions based on their findings.

- **Data Scientist: The Sexiest Job of the 21st Century – Harvard Business Review**

Just to find a good JOB!

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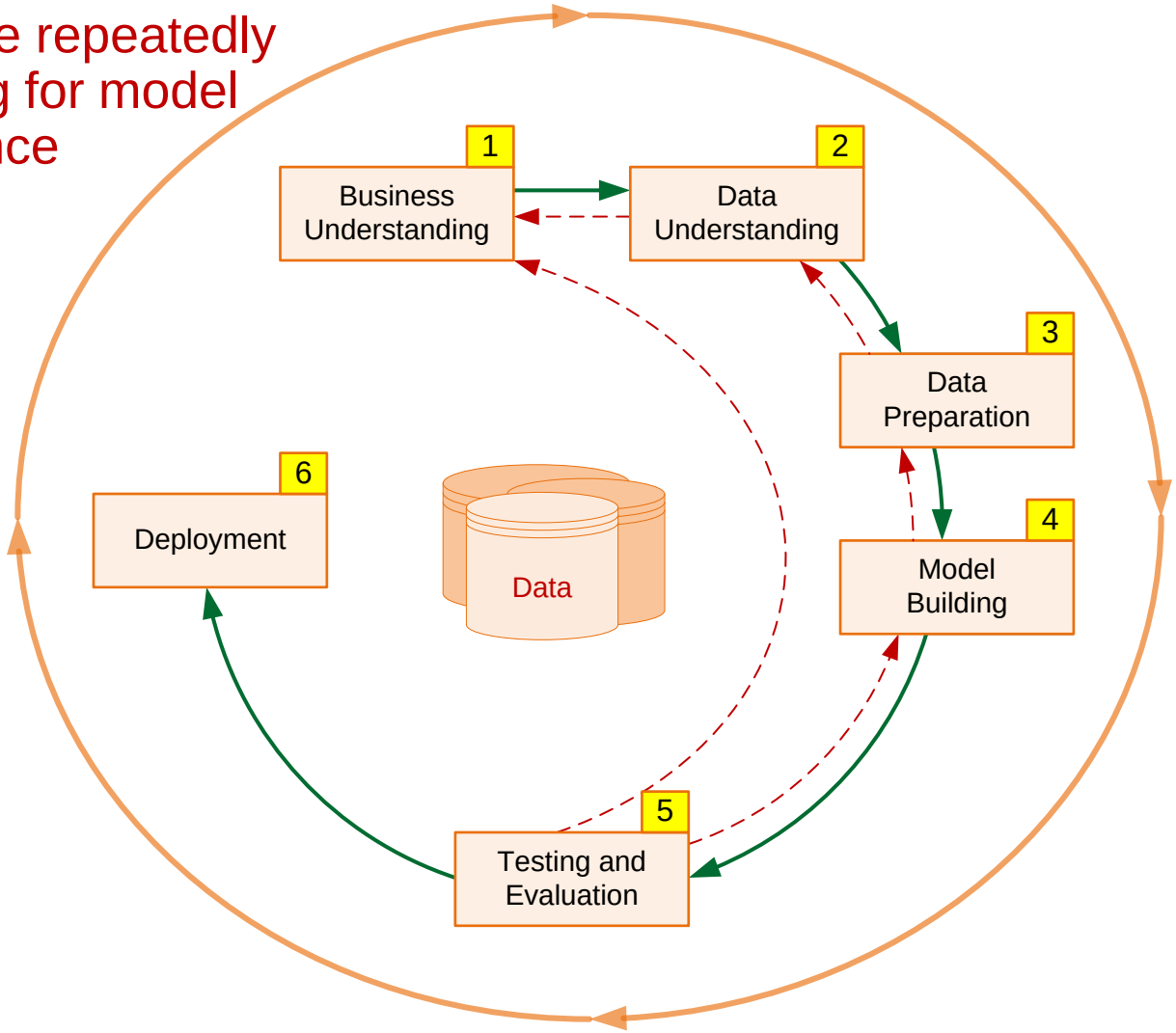
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Reading Material:

- ❑ **DM Chapter 1**

BDA Process

Steps can be repeatedly backtracking for model & performance refinement



BDA Process

- **Cr**oss **I**ndustry **S**tandard **P**rocess for **D**ata **M**ining
- Proposed in 1990s by a European consortium
- Composed of six consecutive phases:
  - Step 1: Business Understanding
  - Step 2: Data Understanding
  - Step 3: Data Preparation
  - Step 4: Model Building
  - Step 5: Testing and Evaluation
  - Step 6: Deployment

Accounts for ~80% of total project time

Your group project may follow this process (except deployment if not applicable)

Course Structure Overview

Advanced Machine Learning Techniques

- Predictive Analytics — Week 2
- Feature Engineering — Week 3
- Clustering Models — Week 4
- Prediction Models — Week 5-8

Neural Networks and Deep Learning

- Artificial Neural Network — Week 9
- Recurrent NN — Week 9
- Image Mining — Week 10
- GenAI and LLM — Week 11

End-to-end Business Applications

Feature Engineering



Feature is the element that you use to describe an object

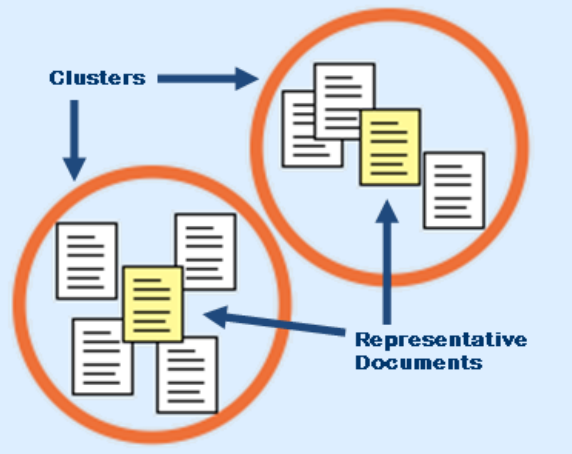


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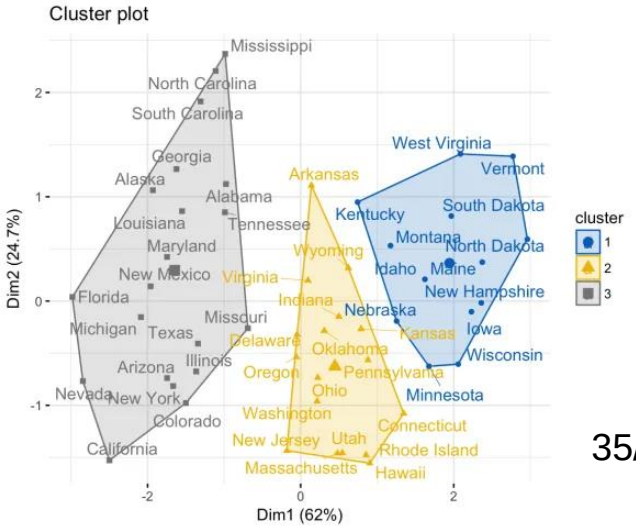
See also www.cs.gmu.edu/~jessica/DimReducDanger.htm

Clustering

- **Understanding**
  - Group related documents for browsing, group genes and proteins that have similar functionality, or group stocks with similar price fluctuations



- **Summarization**
  - Reduce the size of large data sets



Clustering

- Clustering Algorithms

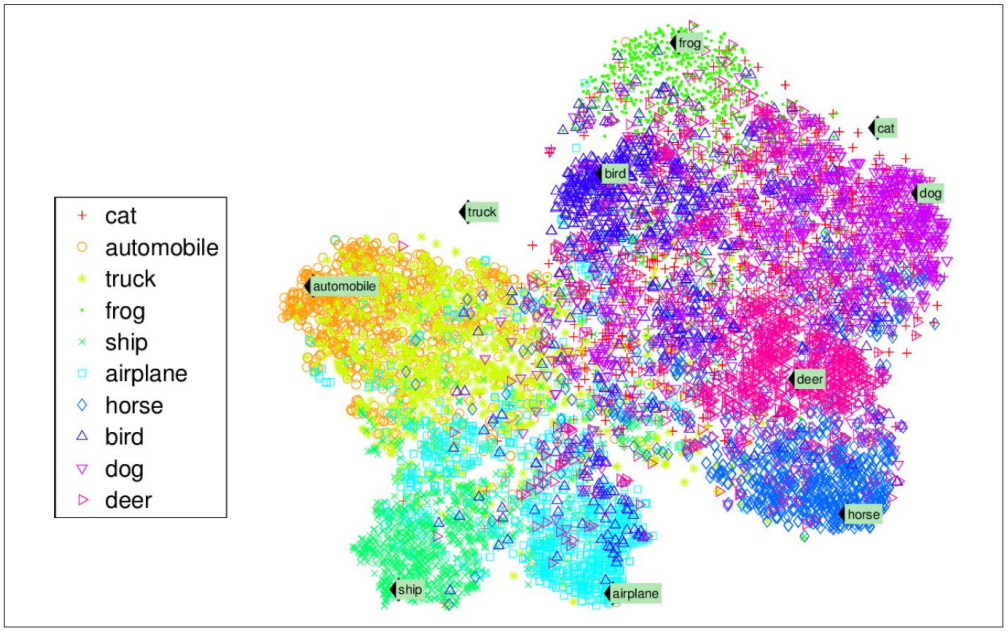
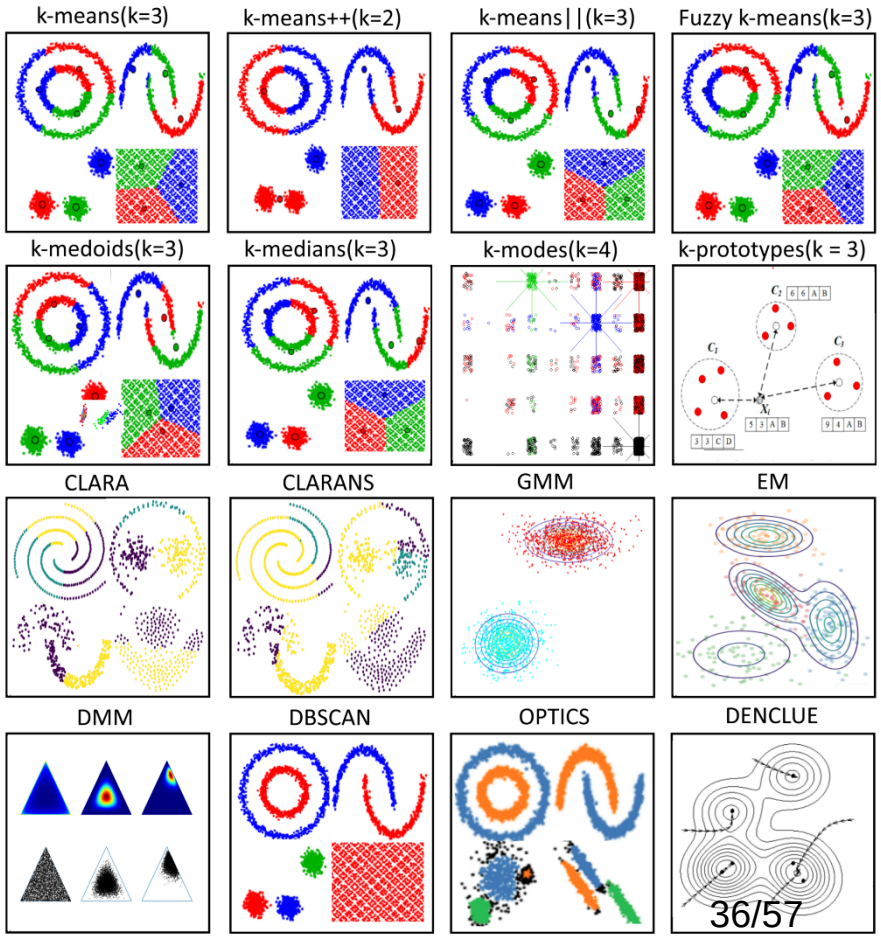
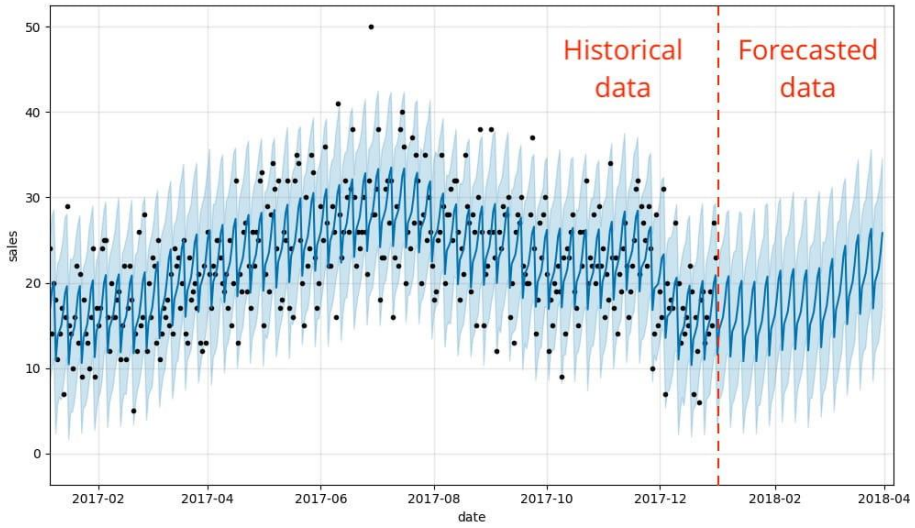


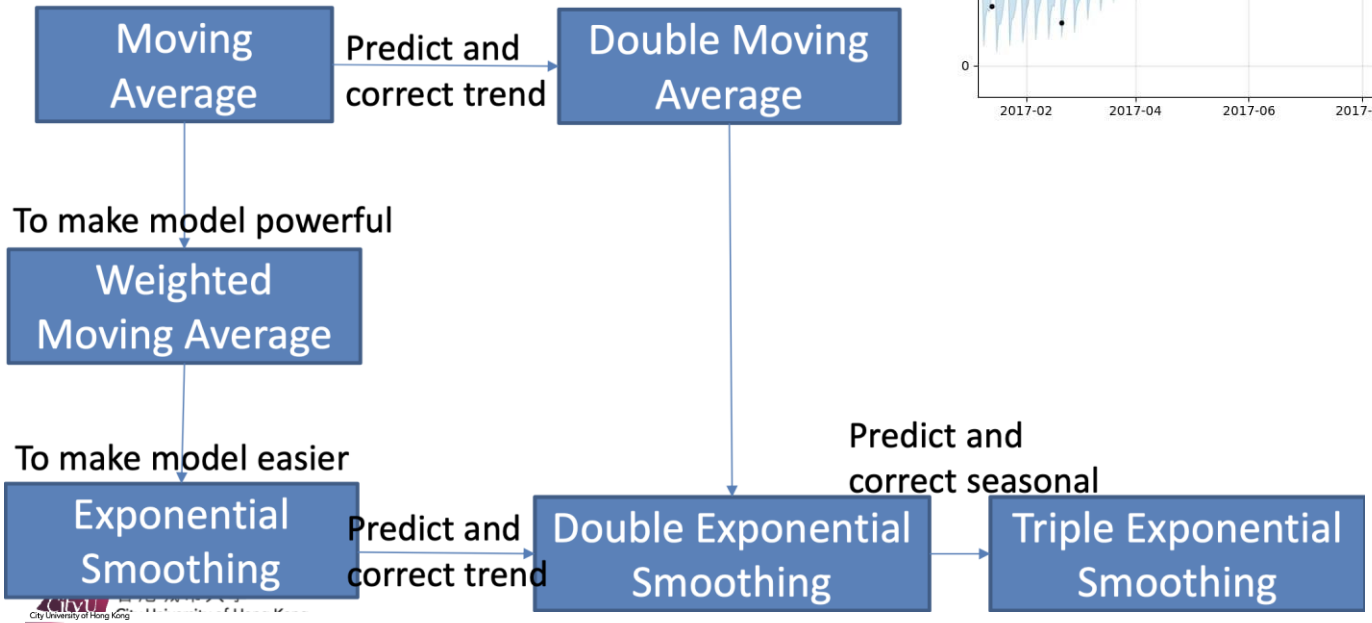
Figure 1-9. Example of a t-SNE visualization highlighting semantic clusters[3]



Predictive Analytics

Time series forecasting





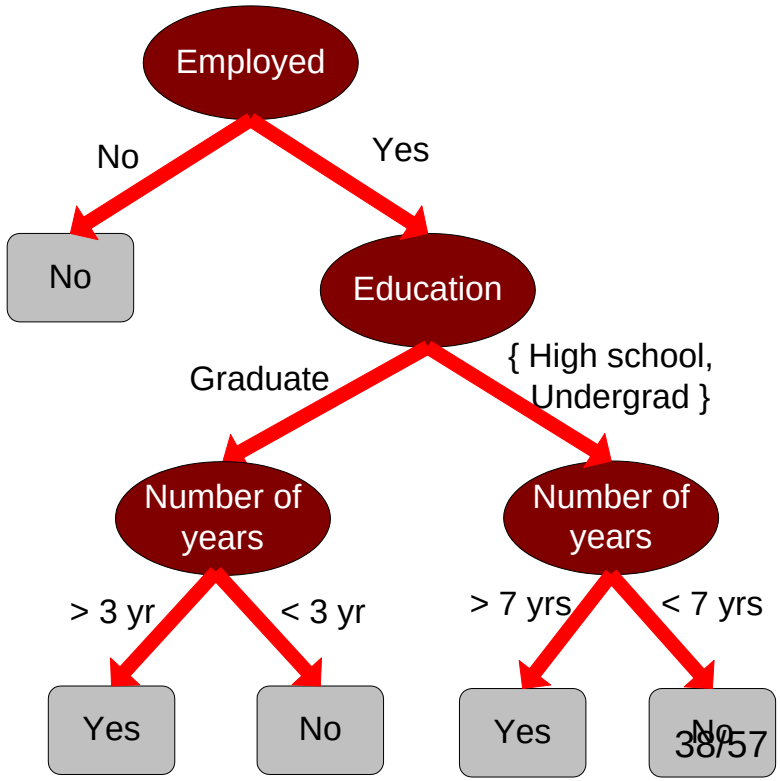
Classification

- Find a model for class attribute as a function of the values of other attributes

Class

Tid	Employed	Level of Education	# years at present address	Credit Worthy
1	Yes	Graduate	5	Yes
2	Yes	High School	2	No
3	No	Undergrad	1	No
4	Yes	High School	10	Yes
…	…	…	…	…

Model for predicting credit worthiness



Advanced Classification

- Ensemble Learning to solve the same problem

				Class
Tid	Employed	Level of Education	# years at present address	Credit Worthy
1	Yes	Graduate	5	Yes
2	Yes	High School	2	No
3	No	Undergrad	1	No
4	Yes	High School	10	Yes
…	…	…	…	…

Model for predicting credit worthiness

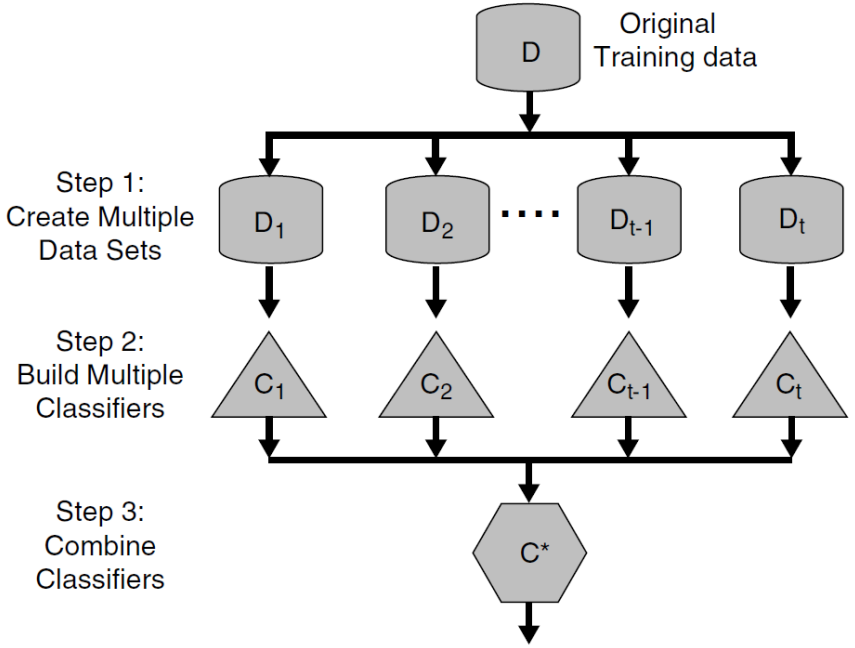


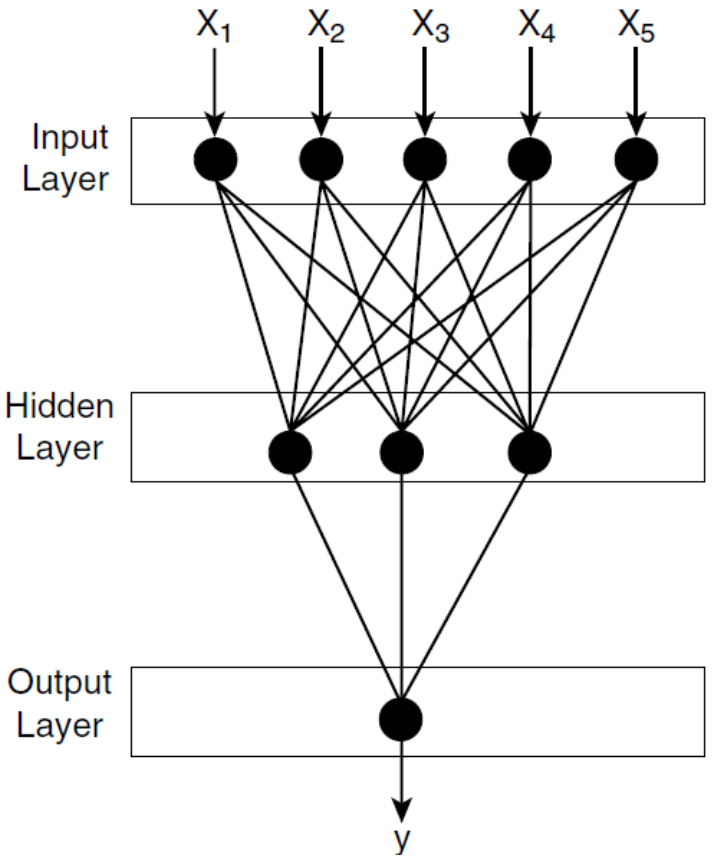
Figure 5.31. A logical view of the ensemble learning method.

Deep Learning

- Neural Network

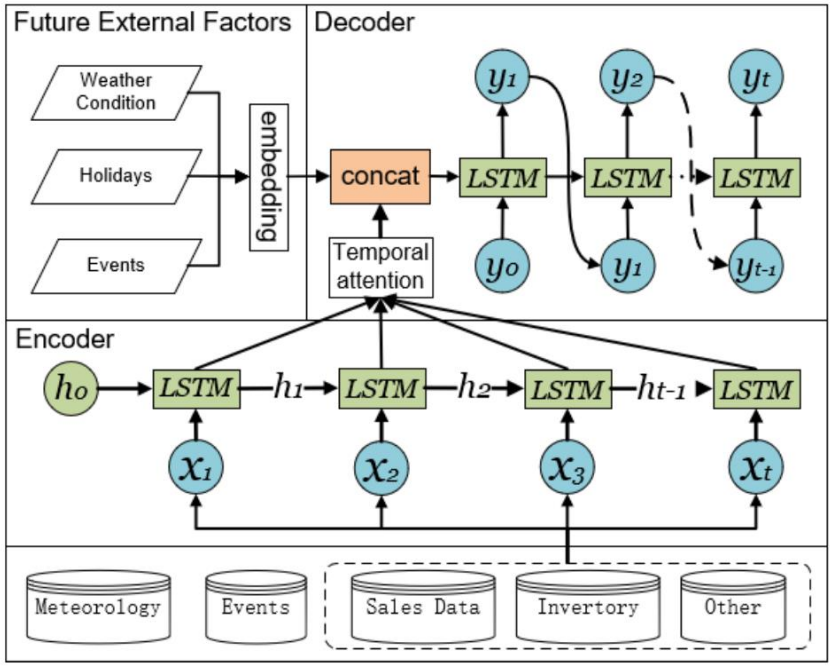
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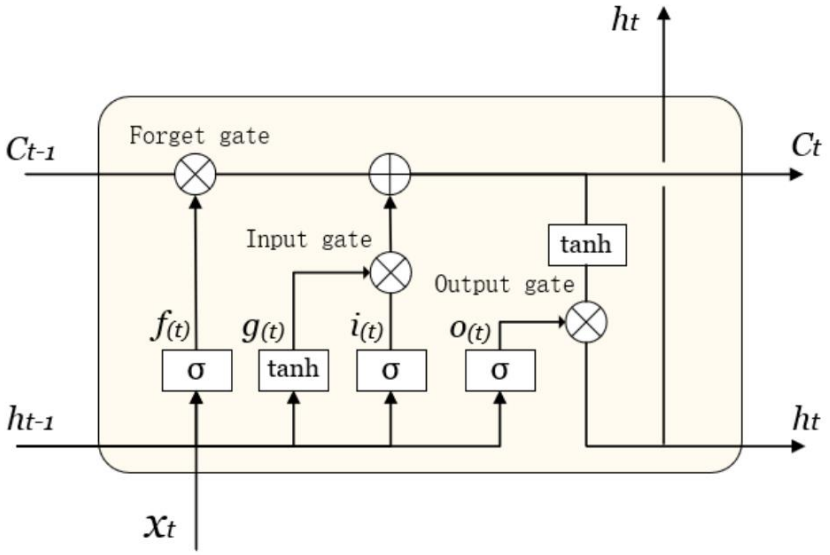


Neural Network

- Sequential Model



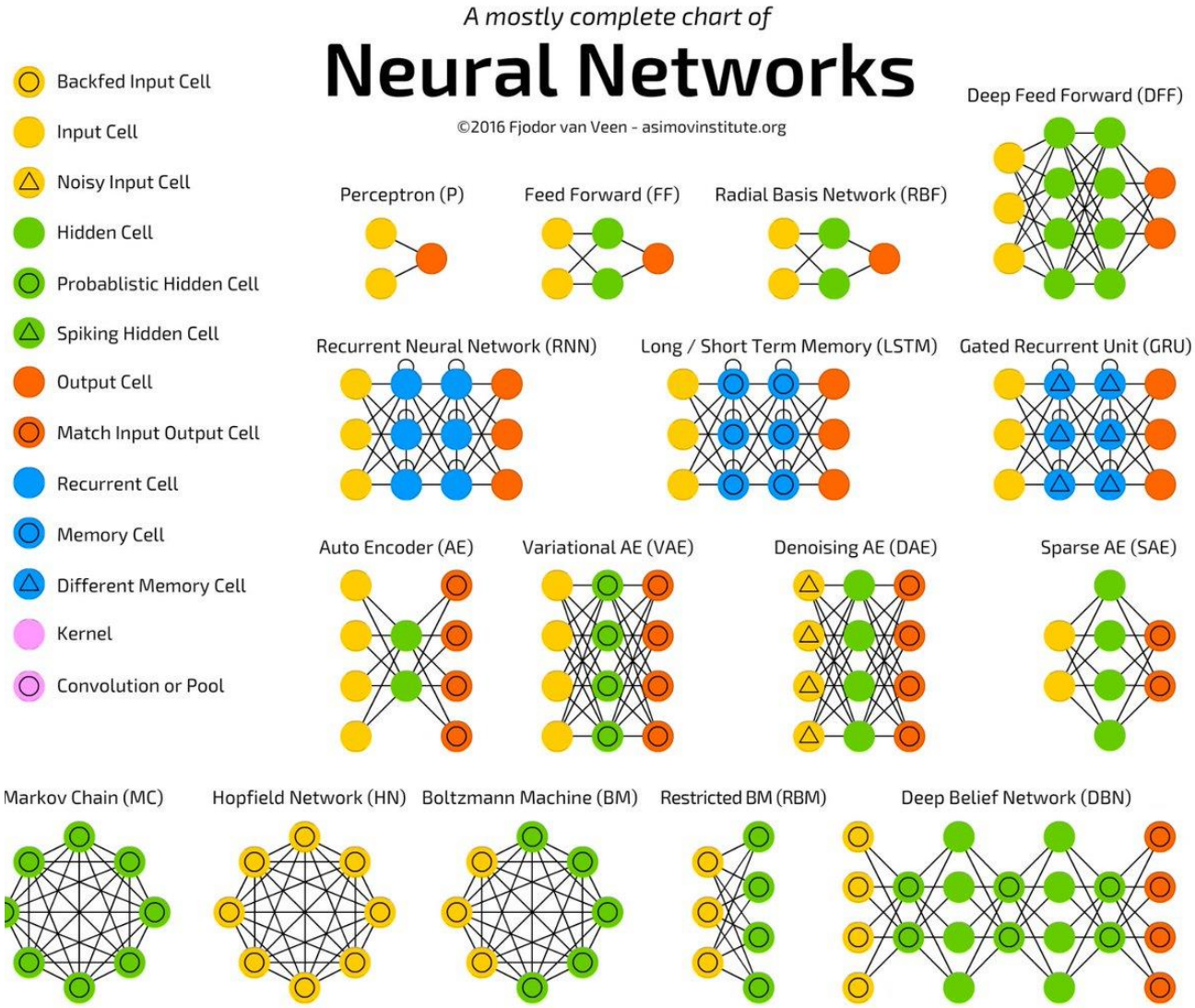
(a) Sequential Model



(b) LSTM Cell

Figure 2: Framework of Deep Sequential Model for Component Consumption Prediction

Neural Network Class



Deep Learning and GenAI

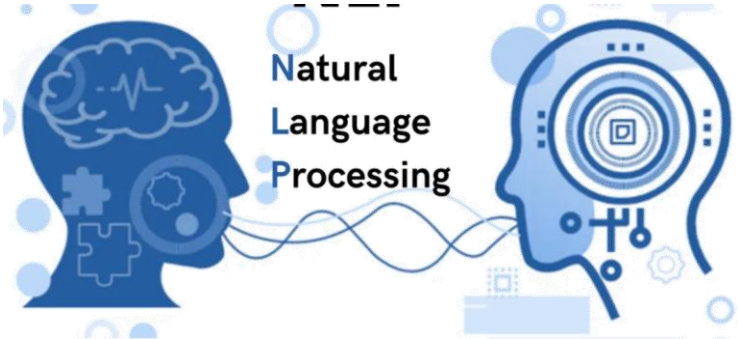
Deep Learning
(Methodological/technical framework)

Prediction, Classification → Generative AI

Extend to generate new content:
text, image, video

Application: **Natural Language Processing (NLP)**

Enabling computers to understand, process and generate natural language.



Large Language Model

- Language Model

  Models capable of generating, understanding, or manipulating text or speech in natural language

  - Neural network-based models
  - Transformer models
    - Attention: Instead of reading word by word, it can examine multiple words at once and understand their relationships

- Large Language Model
  - Large model using Transformer
  - Trained on large-scale text corpora
  - e.g. Generative pre-trained transformer (GPT)

LLM Applications



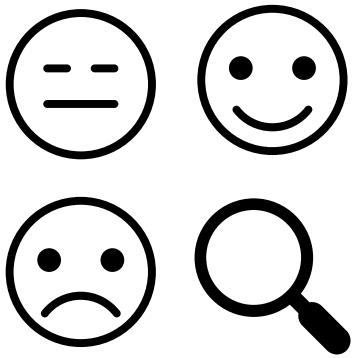
Chatbot

Text

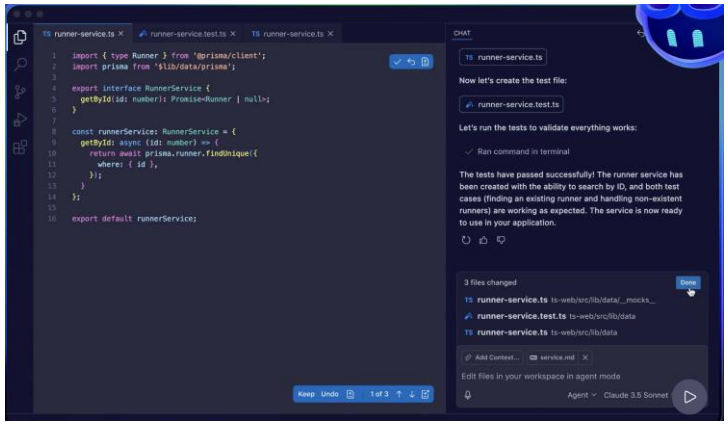
Image

Video

Content generation



Sentiment analysis



Code development

- Translation
- Recommendation
- …

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Anacoda - Jupyternotebook

- https://www.anaconda.com/

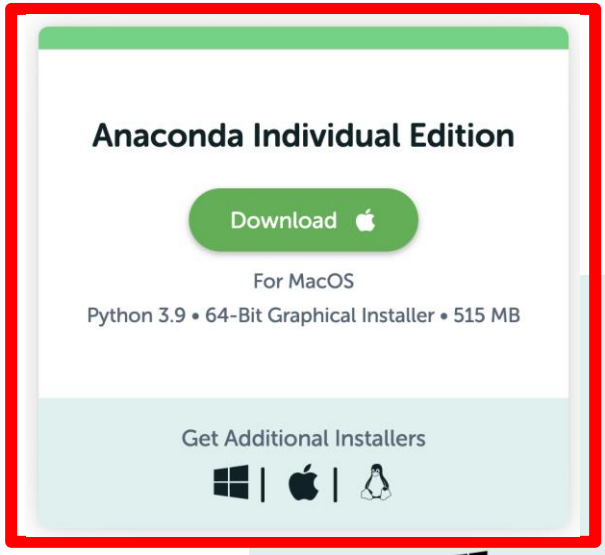


Anaconda was named a May 2019 Gartner Peer Insights Customers' Choice for Data Science and Machine Learning Platforms.



Anacoda - Jupyternotebook

- https://www.anaconda.com/products/individual



Anaconda Installers

Windows

Python 3.9
64-Bit Graphical Installer (510 MB)
32-Bit Graphical Installer (404 MB)

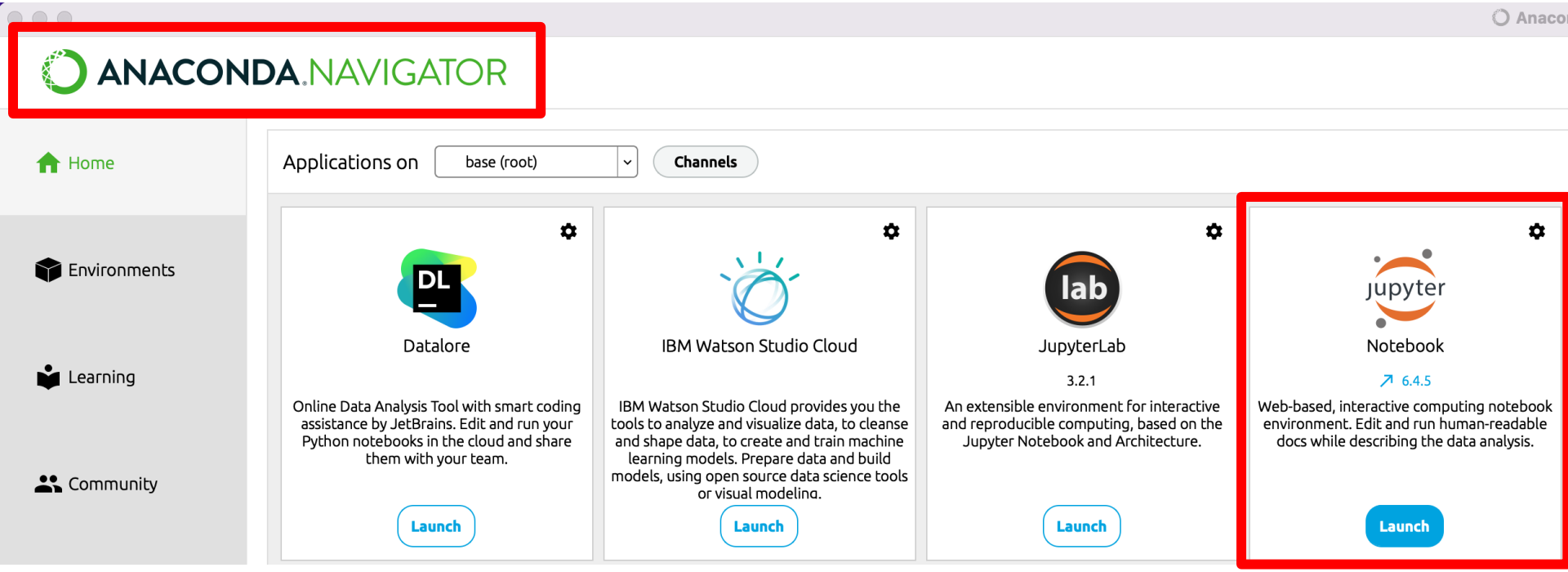
MacOS

Python 3.9
64-Bit Graphical Installer (515 MB)
64-Bit Command Line Installer (508 MB)

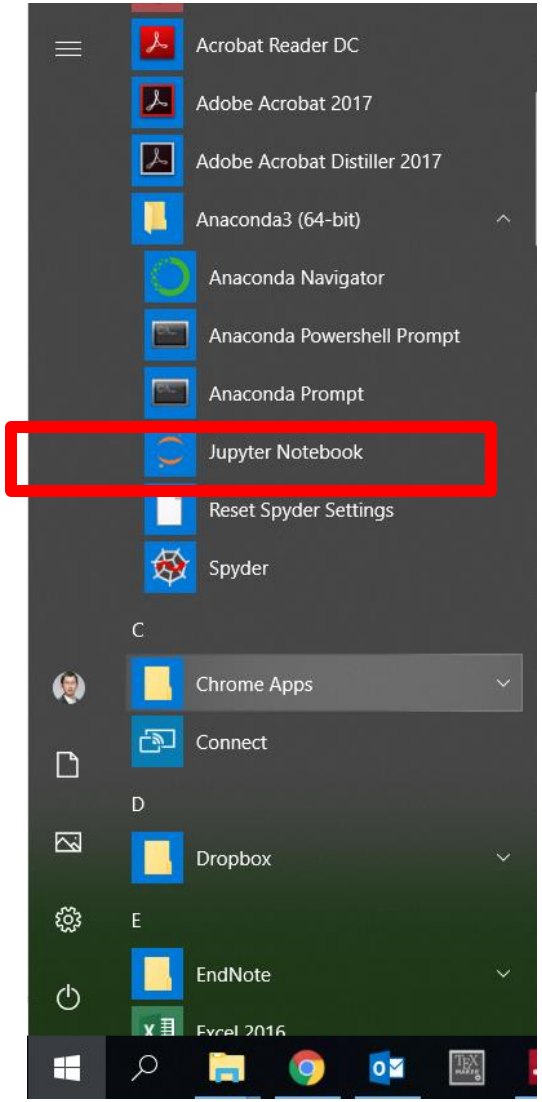
Linux

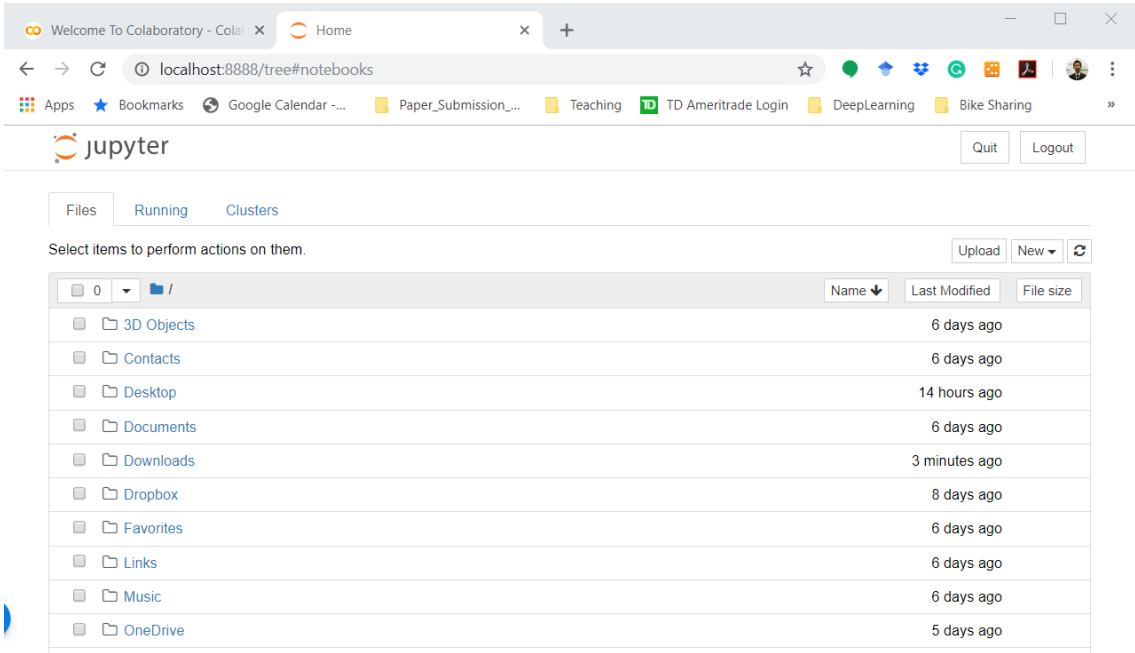
Python 3.9
64-Bit (x86) Installer (581 MB)
64-Bit (Power8 and Power9) Installer (255 MB)
64-Bit (AWS Graviton2 / ARM64) Installer (488 M)
64-bit (Linux on IBM Z & LinuxONE) Installer (242 M)

Anaconda – Jupyter notebook

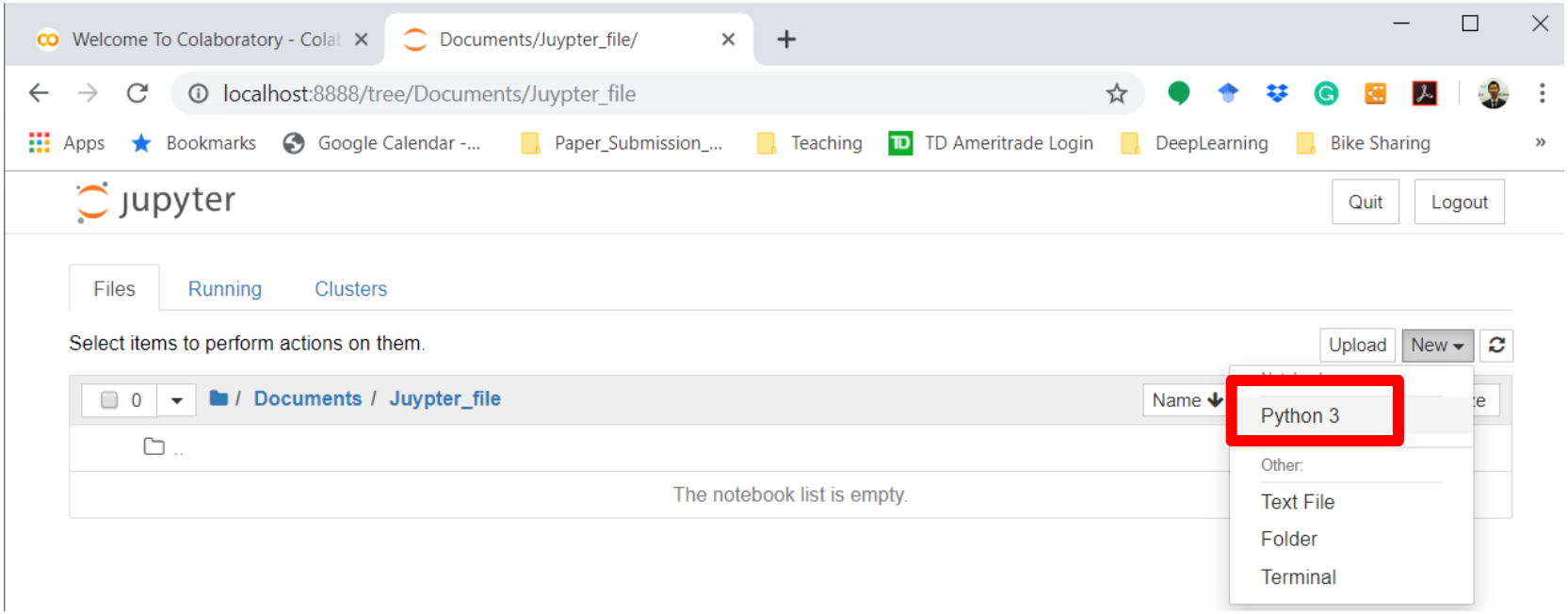


Jupyter notebook

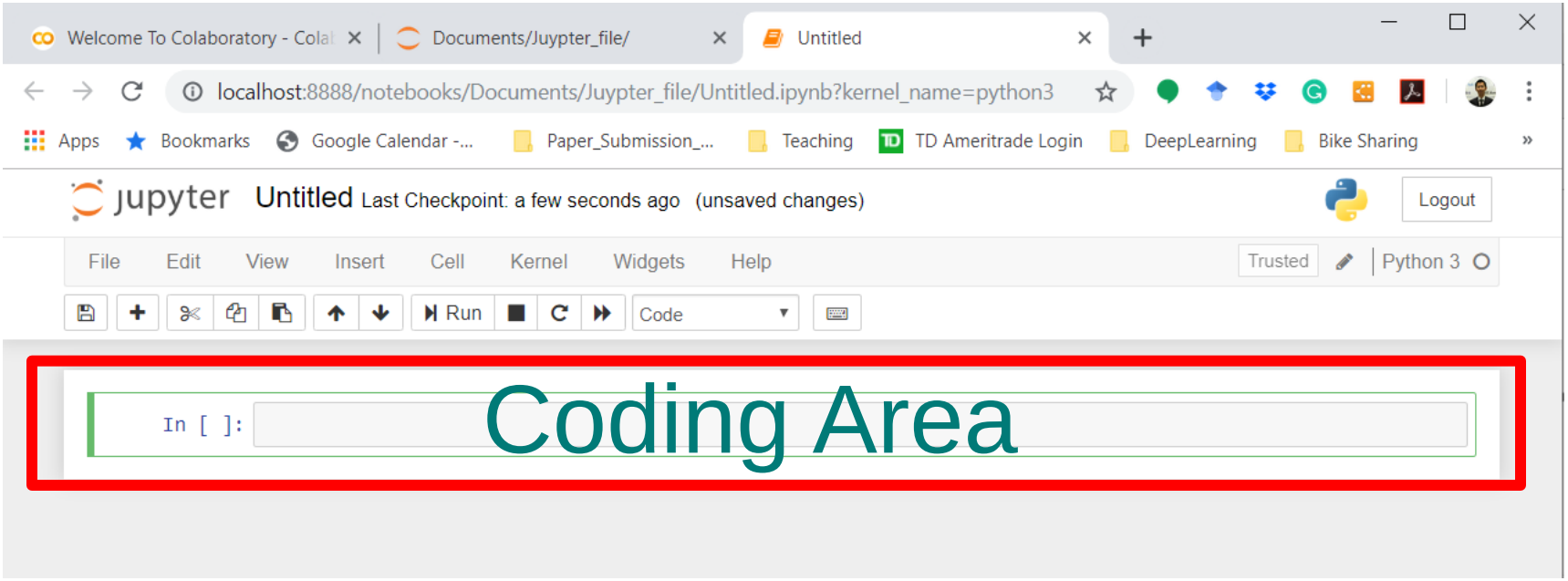




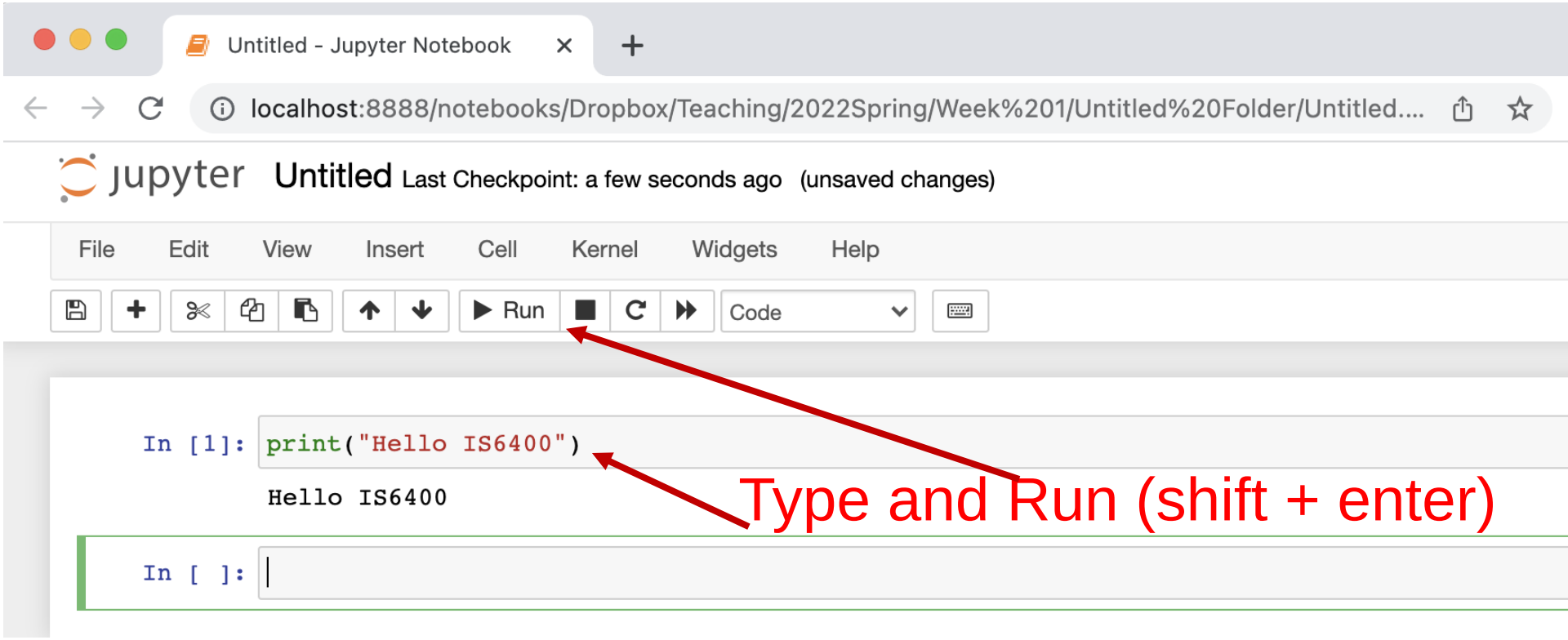
Juypter notebook



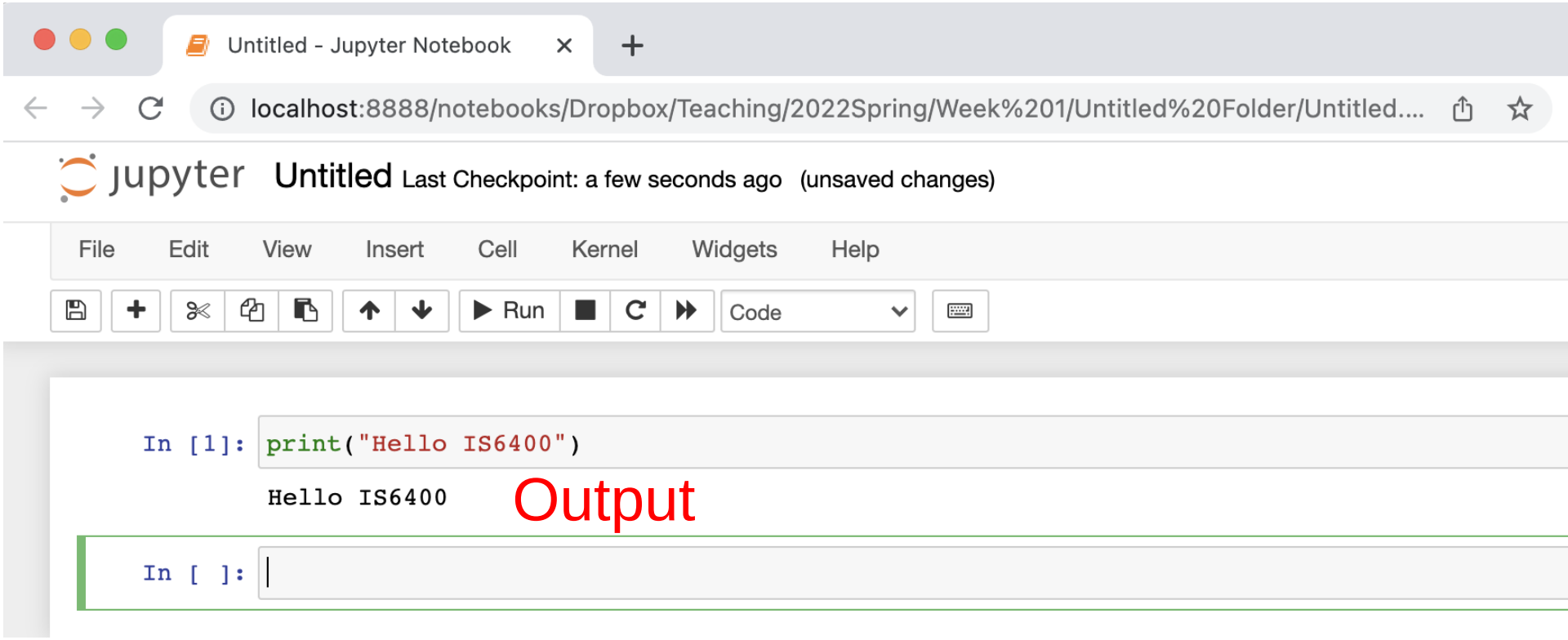
Juypter notebook



Jupyter notebook

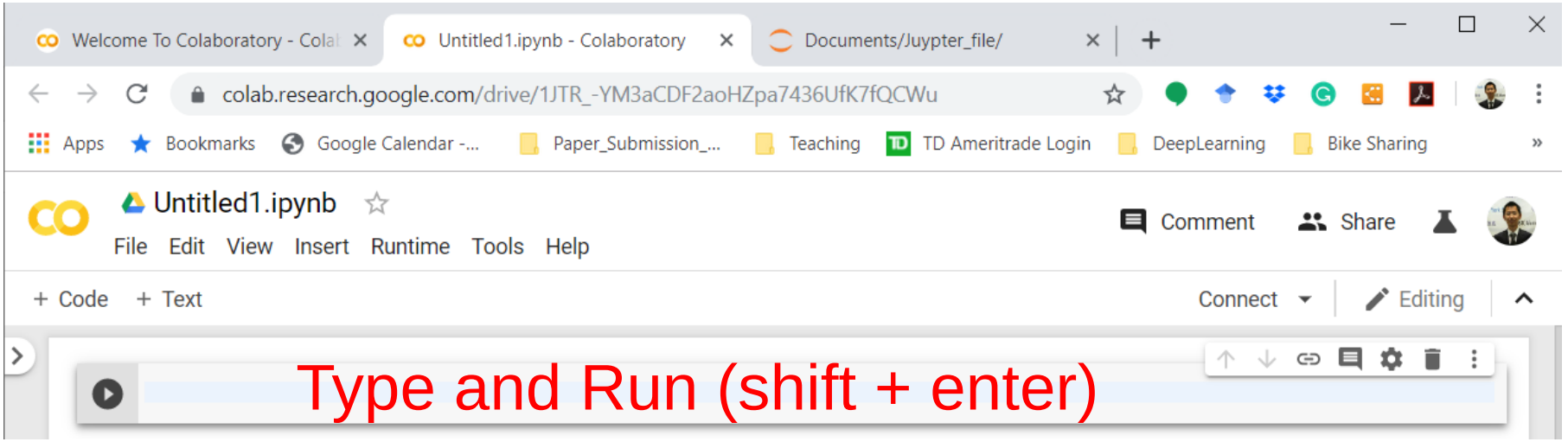


Jupyter notebook



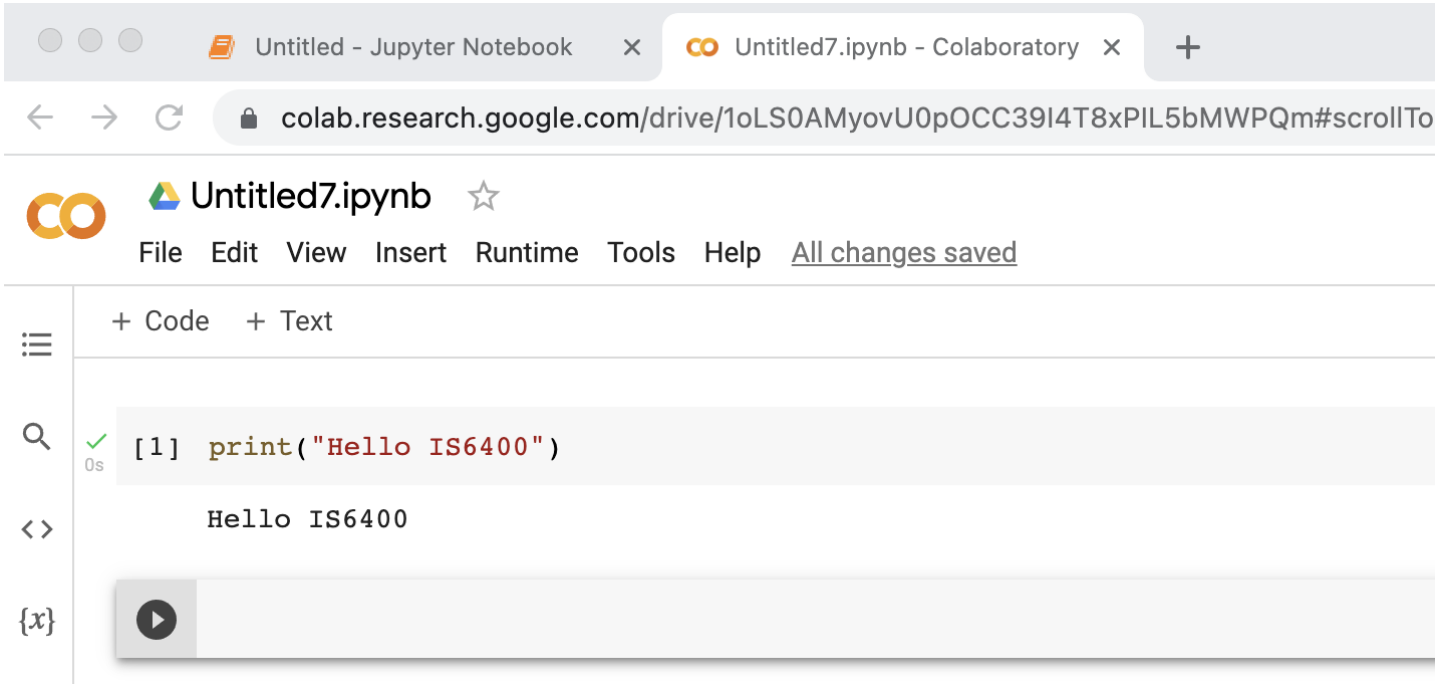
Google Colab

- https://colab.research.google.com/



Google Colab

- https://colab.research.google.com/



Google Colab



Use my cell phone to code?
YES!